B E N C H M A R K R E P O R T

Correct Answer, Wrong Credit

When a Thinking Model Knows the Answer But the Parser Doesn’t See It

Behavioral Analysis of 22 LLMs on a Pump Fault Diagnosis Question

22	77.3%	81.8%	1
LLMs Tested	Reported Pass Rate	Adjusted Pass Rate	Parsing Casualty

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1. Executive Summary

I posed a pump fault diagnosis question to 22 large language models: given five symptoms of a failing submersible pump, identify the most likely root cause from four options. The correct answer is **A) Impeller clogged with debris.**

The reported pass rate is 77.3% (17/22). But the real story of this benchmark lies beneath that number. One model — DeepSeek R1 — answered correctly but was marked as failed due to a parsing issue in the benchmark code. Adjusting for this, the effective pass rate is 81.8% (18/22), with only 4 models genuinely answering incorrectly.

F1 — The Parsing Discovery: DeepSeek R1 produced a correct answer (A) with a thorough explanation, but its response began with a <think> block. The benchmark code evaluated the first character of the raw response ("<") rather than the visible answer. This is a benchmark design lesson: thinking models require adapted parsing. We document the issue and the fix transparently in this report.

F2 — R1's Internal Engineering Debate: Inside its 21,147-character think block, R1 followed a remarkable path: it initially favored A, then rejected it, pivoted to C (electrical winding damage), wrote the pump power equation, reconsidered, and ultimately returned to A. The models that chose C never made it back.

F3 — The "C" Trap: Three models chose C (electrical winding damage) using the same flawed shortcut: "high current + overheating = electrical problem." This mirrors the aerobic trap observed in the Biogas benchmark — a reasoning path that appears logical but misses the mechanical reality.

F4 — The Gemma Scaling Threshold: The Gemma family shows a clear size-performance pattern on this question: 1B (fail, zero reasoning), 4B (fail, wrong answer), 12B (fail, wrong answer), 27B (pass). A clean step function that was not observed in the earlier Material Selection benchmark where all four passed.

F5 — All Failures Are Google: The 4 genuinely incorrect models are all from Google's portfolio: Gemma 3 1B, Gemma 3 4B, Gemma 3 12B, and Gemini 2.0 Flash. Claude (5/5), Qwen (4/4), and DeepSeek (3/3 on knowledge) had no failures.

2. The Question

The test presents a realistic fault diagnosis scenario. A submersible pump in a wastewater treatment plant exhibits five specific symptoms. The operator must identify the root cause from four options.

Table 1: Symptom Profile

#	Symptom	Diagnostic Value
1	Submersible pump motor is overheating	High load indicator
2	Pump flow rate decreased by 40%	Restriction or loss of capacity
3	Ammeter shows current 30% above normal	Motor under excess load
4	**No unusual sounds from the pump**	**KEY — rules out B and D**
5	**Discharge pressure is normal**	**KEY — rules out C**

Table 1: Symptoms 4 and 5 are the critical differentiators. Symptom 4 (no sounds) eliminates bearing failure and cavitation. Symptom 5 (normal pressure) challenges electrical winding damage as the cause.

Table 2: Answer Options

	Cause	Ruled Out By	Assessment
A	**Impeller clogged with debris**	—	**CORRECT — explains all 5 symptoms**
B	Bearing failure	Symptom 4 (no sounds)	Would produce grinding/squealing
C	Electrical winding damage	Symptom 5 (normal pressure)	THE TRAP — explains 1–3 but not 5
D	Cavitation	Symptom 4 (no sounds)	Would produce crackling/popping

Table 2: Option C is the deliberate trap. It fits symptoms 1–3 (overheating, flow drop, high current) but fails to explain symptom 5 (normal discharge pressure).

The evaluation asserts that the first character of the response must be “A.” This simple parser, as we discovered, does not account for thinking models that prepend <think> blocks before their visible answer.

3. Results

Table 3: Answer Distribution

Answer	Count	Status	Models
A	17 + 1*	CORRECT	All Claude, Qwen, DeepSeek; most Gemini; Gemma 27B
C	3	WRONG — the “C” trap	Gemma 3 1B, Gemma 3 12B, Gemini 2.0 Flash
B	1	WRONG	Gemma 3 4B

*Table 3: *DeepSeek R1 answered A correctly but was marked as failed due to <think> block parsing. See Section 4.1.*

Reported vs. Adjusted Results: The benchmark code reported 17/22 pass (77.3%). After discovering that DeepSeek R1’s visible response correctly starts with “A,” the adjusted knowledge-based pass rate is 18/22 (81.8%). Both numbers are presented throughout this report for transparency.

Table 4: Complete Results — All 22 Models

Model	Answer	Status	Out Tok	Latency	Family
Claude Haiku 4.5	A	PASS	249	3.9s	Claude
Claude Opus 4.1	A	PASS	317	12.8s	Claude
Claude Opus 4.5	A	PASS	314	9.3s	Claude
Claude Sonnet 4	A	PASS	321	9.4s	Claude
Claude Sonnet 4.5	A	PASS	308	9.0s	Claude
DeepSeek V3.1	A	PASS	454	3.8s	DeepSeek
DeepSeek V3.2	A	PASS	156	7.3s	DeepSeek
Gemini 2.0 Flash Lite	A	PASS	321	3.2s	Gemini
Gemini 2.5 Flash	A	PASS	309	11.0s	Gemini
Gemini 2.5 Pro	A	PASS	359	20.5s	Gemini
Gemini 3 Flash Preview	A	PASS	355	8.0s	Gemini
Gemini 3 Pro Preview	A	PASS	348	68.3s	Gemini
Gemma 3 27B	A	PASS	380	10.4s	Gemma
Qwen3 235B A22B Instruct	A	PASS	167	2.1s	Qwen
Qwen3 Coder 480B	A	PASS	254	2.5s	Qwen
Qwen3 Next 80B Instruct	A	PASS	260	2.4s	Qwen
Qwen3 Next 80B Thinking	A	PASS	10,492	56.1s	Qwen
DeepSeek R1 (CoT)	A*	PARSE*	4634	30.5s	DeepSeek
Gemini 2.0 Flash	C	FAIL	345	3.8s	Gemini
Gemma 3 12B	C	FAIL	210	8.5s	Gemma
Gemma 3 4B	B	FAIL	320	7.7s	Gemma

Gemma 3 1B	**C**	**FAIL**	1	1.0s	Gemma

Table 4: All 22 models sorted by pass/fail, then family. PARSE indicates correct knowledge but failed due to parsing. A* indicates the visible answer is A but the raw response starts with <think>.*

4. LLM Behavior Analysis

4.1 The Parsing Discovery — When Code Meets Thinking Models

This section documents a benchmark design issue discovered during analysis. We present it transparently as both a finding and a lesson for benchmark authors working with thinking models.

The benchmark code evaluates answers with a simple parser: response.strip()[0].upper(). This extracts the first character of the raw API response and checks if it equals “A.” For 21 of 22 models, this works correctly. For DeepSeek R1, it does not.

R1’s raw response begins with a <think> block — an internal reasoning trace that precedes the visible answer. The first character is therefore “<” (the opening of the XML tag), not “A.” The visible portion of R1’s response, which is what a human user would see, clearly starts with “A” followed by a well-structured explanation.

Table 5: The Parsing Issue

	Current Code	Recommended Fix
Logic	response.strip()[0].upper()	Strip <think> block, then check [0]
R1 raw input	"<think>...21K chars...</think>\nA\n..."	"A\nThe most likely root cause is A)..."
First char detected	**"<" → FAIL**	**"A" → PASS**
Python fix	first_char = answer[0].upper()	clean = re.sub(r’<think>.*?</think>’, ’’, resp).strip()

Table 5: The parsing issue and recommended fix. This affects any benchmark evaluating thinking models that emit <think> blocks.

This is worth noting for the broader benchmark community: as thinking models become more prevalent, benchmark parsers that assume the first character is the answer may need to be adapted. The fix is straightforward (strip known reasoning tags before evaluation), but the failure mode is silent — the model appears to have answered incorrectly when it has not.

4.2 R1's Internal Journey — 21,147 Characters of Engineering Debate

DeepSeek R1's think block on this question is remarkable. In 85 paragraphs and over 21,000 characters, R1 conducted a genuine engineering analysis — including writing the pump power equation, debating pump curves, and changing its mind multiple times. The decision flow reveals how extended reasoning can both endanger and ultimately rescue a correct answer.

Table 6: R1's Decision Flow (reconstructed from <think> block)

Phase	Internal Reasoning	Leaning	Status
1–10	Analyzes all 4 options. A fits most symptoms. But questions whether clogged impeller explains normal pressure.	A	Initial hypothesis
11–30	Writes pump power equation: P = Q×H×ρ×g / η. Calculates that reduced flow should mean less power, not more. Concludes A doesn't explain high current.	A → ?	Doubt enters
47	*"This suggests that the clogged impeller is not the cause."*	✗ A	A rejected
51	*"Therefore, the most likely cause is electrical winding damage (C)."*	C	Wrong answer chosen
52–63	Continues analyzing. Questions whether winding damage explains normal pressure. Considers phase imbalance, slip speed, system curves.	C → ?	Doubt returns
64	*"But consider: if the impeller is clogged, the pump might be operating at very low efficiency, and also the motor might be overloaded because the clog causes mechanical resistance."*	A?	Reconsideration
76	*"Given the options, I think the best fit is A) Impeller clogged with debris."*	A	Return to correct
Final	"So, after reconsideration, I choose A." Visible response begins with A and provides structured explanation.	A ✓	Correct answer

Table 6: R1's internal decision flow reconstructed from the 85-paragraph think block. The model traversed A → reject A → choose C → doubt C → return to A. Gemini 2.0 Flash and Gemma 12B reached the "choose C" stage but never returned.

> *This suggests that the clogged impeller is not the cause.*
> — DeepSeek R1, <think> block, paragraph 47 (moments before choosing C)

> *Given the options, I think the best fit is A) Impeller clogged with debris.*
> — DeepSeek R1, <think> block, paragraph 76 (after 29 more paragraphs of analysis)

The Biogas comparison is illuminating: in the Biogas Desulfurization benchmark, R1 also considered the correct answer (E) multiple times but ultimately chose wrong (D). Here, the extended reasoning saved R1 — it found its way back to A after a deep detour through C. This suggests that CoT's value may depend on whether the correct answer can withstand rigorous cross-examination. When the right answer is robust (as A is here), extended reasoning can validate it. When it is subtle (as E was in Biogas), extended reasoning may erode confidence in it.

4.3 The "C" Trap — When Electrical Thinking Overrides Mechanical Reality

Three models independently constructed the same flawed argument for choosing C (electrical winding damage). The reasoning chain is consistent:

Table 7: The Common Reasoning Chain in C-Choosing Models

Step	Reasoning	Error?
1	Motor is overheating and drawing high current	Correct observation
2	High current + overheating = electrical problem	**THE SHORTCUT — ignores mechanical causes**
3	Winding damage causes shorts → more current → heat	Plausible but incomplete
4	Dismiss A: "clogged impeller wouldn't cause this much current"	**Critical error — it would**
5	Ignore or rationalize symptom 5 (normal discharge pressure)	Winding damage should reduce pressure

This parallels the "aerobic trap" from the Biogas benchmark: in both cases, models followed a reasoning shortcut that was biochemically or electrically defensible in isolation but failed when confronted with the full symptom picture. In Biogas, "SOB = aerobic → aeration basin." Here, "high current = electrical → winding damage." Both traps share a common structure: the model anchors on the most salient symptom and reasons from there, rather than using all symptoms as a constraint set.

4.4 Gemma Scaling — A Clear Size-Performance Threshold

Table 8: Gemma Family Scaling Pattern

Model	Answer	Status	Output	Reasoning Quality
Gemma 3 1B	C	**FAIL**	1 token	No reasoning at all — single letter
Gemma 3 4B	B	**FAIL**	~320 tok	Elimination reasoning present but wrong conclusion
Gemma 3 12B	C	**FAIL**	~210 tok	Falls into C trap — electrical shortcut
Gemma 3 27B	A	**PASS**	~380 tok	Full elimination + wastewater context

Table 8: A step function at 27B. Below this threshold, no Gemma model answered correctly. This pattern was not present in the Material Selection benchmark where all four Gemma models passed.

This is the clearest scaling signal across the three WWTP benchmarks so far. In the Material Selection benchmark (easy), all four Gemma models passed. Here (medium difficulty), only 27B passes. In the Biogas benchmark (hard), we would need to check if 27B also failed. If it did, this would suggest a consistent difficulty threshold that shifts upward with model size — though more data points would be needed to confirm such a pattern.

4.5 Thinking Models — When Extended Reasoning Helps (and When It Costs)

Table 9: Thinking vs. Standard Model Comparison

Model	Output Tok	Latency	Answer	Status	Type
DeepSeek V3.2	156	7.3s	A	PASS	Standard
DeepSeek R1 (CoT)	**4,634**	30.5s	A*	PARSE*	Thinking
Qwen3 Next 80B Instruct	260	2.4s	A	PASS	Standard
Qwen3 Next 80B Thinking	**10,492**	56.1s	A	PASS	Thinking

Qwen3 Thinking used 40x more tokens than its Instruct counterpart (10,492 vs 260) for the same correct answer. DeepSeek R1 used 30x more than V3.2 (4,634 vs 156). Both thinking models arrived at the correct answer, but the resource cost is striking.

However, R1's extended reasoning arguably added value here — it traversed the C trap and found its way back. Had R1 been a non-thinking model, it might have stopped at paragraph 51 ("therefore C") like Gemini Flash did. This is the first benchmark in our series where CoT reasoning appears to have rescued a model from a wrong answer, even if the rescue was invisible to the parser.

4.6 Wastewater Vocabulary — Domain Terms as a Signal

Table 10: Domain-Specific Terms in Model Responses

Term	In Passed (18)	In Failed (4)	Significance
"rags" or "wipes"	10	0	Field-specific — common impeller clog material
"wastewater" context	10	0	Situates answer in operational reality
"ragging" (technical term)	2	0	Expert vocabulary — industry-specific
"volute" (pump component)	2	0	Demonstrates pump anatomy knowledge
"grinding" / "squealing" sounds	15	2	Used to rule out bearing failure

Table 10: Domain-specific vocabulary appears almost exclusively in correct responses. This mirrors the Biogas benchmark finding where terms like "microaerophilic zones" appeared only in passing models.

A recurring pattern across the WWTP benchmark series: models that use field-specific terminology (rags, ragging, volute) tend to answer correctly, while models using only generic technical language tend to fail. This does not prove causation — the vocabulary may simply be a marker of deeper domain knowledge rather than a cause of correct reasoning — but it appears consistently enough to be worth noting.

5. Cross-Benchmark Comparison — Three Questions, One Model Set

Table 11: Three WWTP Benchmarks Side by Side

Dimension	Material Selection	Root Cause Analysis	Biogas Desulfurization
Pass Rate	**100% (22/22)**	**77–82% (17–18/22)**	**14.3% (3/21)**
Difficulty	Easy — binary choice	Medium — 4 options with trap	Hard — 5 options with deep trap
Trap Type	None	"C" trap (electrical)	"D" trap (aerobic)
Claude Family	5/5	5/5	0/4
Qwen Family	4/4	4/4	2/4
Gemma Family	4/4	1/4 (27B only)	0/4
Thinking Models	Passed, high cost	R1 correct but parsed wrong	Both failed
Key Insight	Easy Qs reveal efficiency	Medium Qs reveal reasoning traps	Hard Qs reveal domain depth

Table 11: The three benchmarks form a difficulty gradient that reveals different aspects of model behavior at each level.

Models that passed all three benchmarks: only DeepSeek V3.2, Qwen3 235B A22B Instruct, and Qwen3 Next 80B Instruct passed all available tests. These three models come from two families and have no obvious common trait other than being mid-to-large instruct-tuned models. Notably, no flagship model (Claude Opus, Gemini Pro, DeepSeek R1) achieved a perfect record across the series.

6. Benchmark Design Lessons

The R1 parsing discovery is the most practically useful finding in this report for other benchmark authors. We document three lessons:

Lesson 1 — Parse the visible response, not the raw output. Thinking models prepend internal reasoning blocks that are not part of the user-facing answer. Benchmark code should strip these before evaluation. A simple regex (re.sub(r'<think>.*?</think>', '', response, flags=re.DOTALL)) handles the most common format.

Lesson 2 — Log the raw response for analysis. The think block itself is a rich data source for behavioral analysis. Discarding it for evaluation purposes does not mean discarding it for research purposes. R1's 85-paragraph internal debate would have been invisible without access to the raw output.

Lesson 3 — Report both raw and adjusted results. When a parsing issue is discovered, the transparent approach is to report both numbers. This report presents 77.3% (reported) and 81.8% (adjusted) throughout, allowing readers to draw their own conclusions.

7. Conclusions

This benchmark sits between the easy Material Selection question (100% pass) and the hard Biogas Desulfurization question (14.3% pass). At this medium difficulty level, we observe a new set of behaviors.

The “C trap” caught 3 models with a reasoning shortcut that mirrors the “D trap” in Biogas: in both cases, models anchored on the most prominent symptoms and followed the most common textbook association rather than reasoning from the full constraint set. This pattern appears robust enough across two independent benchmarks to warrant attention.

The parsing discovery is perhaps the most transferable finding: as thinking models proliferate, benchmark infrastructure needs to evolve. A correct answer hidden behind a <think> tag is a new category of failure — not model failure, but evaluation failure.

DeepSeek R1’s internal journey is the most intellectually interesting finding. A model that wrote the pump power equation, changed its mind twice, and ultimately arrived at the correct answer — only to lose credit because of an XML tag — is a story that captures something real about the current state of LLM evaluation.

As the third entry in the WWTP benchmark series, this report strengthens an emerging principle: model performance on industrial tasks cannot be predicted by general capability rankings alone. The three models that passed all three benchmarks (DeepSeek V3.2, Qwen3 235B, Qwen3 Next 80B Instruct) are not the largest, most expensive, or highest-ranked models in any general leaderboard.

8. Methodology Notes

Non-determinism: LLMs produce different outputs on repeated runs. This test used a single run per model. Results should be interpreted as indicative rather than absolute.

Single question: This report analyzes one question from a broader benchmark suite. Family-level and reasoning-pattern conclusions should be validated across the full question set.

API conditions: All models were accessed through the Kaggle Benchmarks platform API with default temperature settings. Different configurations may produce different results.

Parsing disclosure: The benchmark code evaluates response.strip()[0].upper() against the expected answer. This does not handle <think> blocks. The adjusted results presented in this report are based on manual inspection of the visible response text.

Output token counts: Some Gemma models did not report output token counts via the API. Where unavailable, estimates are based on response character length.